From populations to individuals and back again

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Regius Professor of Surgery
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Chair, Precision-Panc Therapeutic Development Platform
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GPOL
GLASGOW PRECISION ONCOLOGY LABORATORY

ARGO Accelerating Research in Genomic Oncology
International Cancer Genome Consortium

CANCER RESEARCH UK | PRECISION PANC

Realising the Promise of Precision Oncology

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On our Planet.

Every year…

9.6 million people with cancer
run out of options…

…and die.

Every year…

Less than 1 patient out of 100 will access new treatments or clinical trials.

Every year…

Almost all clinical trials fail to reach recruitment targets, and most never complete.

Every year.

Most potential treatments are never tested

sit on a shelf

Lost.

And nothing changes.

Year after year.

Trillions of tears.

Tamoxifen, the treatment that has helped dying cancer patients more than all other drug treatments combined

Almost by accident.

Was rescued from a dusty shelf

We have all the drugs we need to double survival for cancer

We test only a small fraction of them,
in a fraction of cancer types

So what is the problem…

Are we the problem.

A combination of total luck, a great oncologist and determination to live has enabled me to access my current trial and get an extra four years of life I didn’t expect to have.

As a result I’ve seen my eldest go to uni, the other three all grow up.

All because I chanced upon a drug that precisely targeted my cancer.

Lesley Stephen
Cancer Patient
Edinburgh, 2019

In just 1 hour of our meeting today…

20 people in the United Kingdom
will die of cancer.

1000 people on our planet.

We need to do something…

What future can we build?

University of Glasgow

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Parachutes have never been proven to work in a randomised controlled trial

“Can you believe that not that long ago we would bleed people and do other barbaric things to treat them? How crazy is that.”

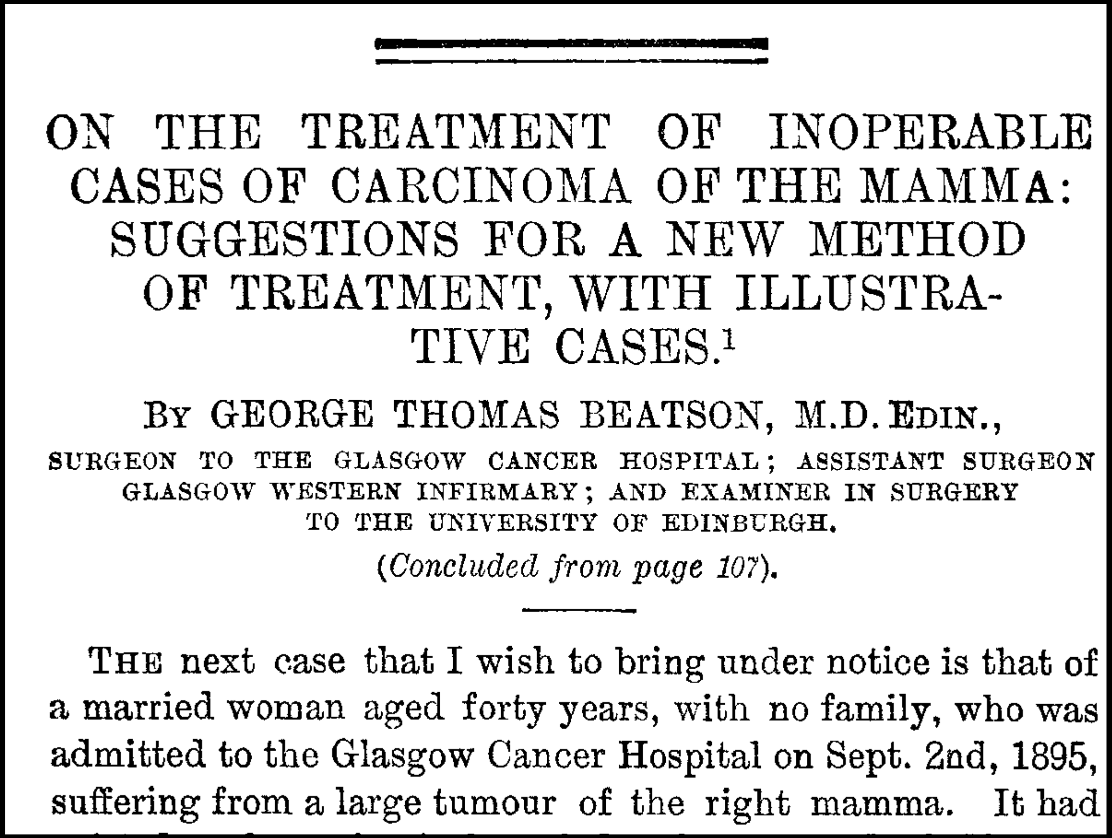
Member of the public 2025

“Can you believe that not that long ago we gave toxic chemicals to people to treat diseases without knowing if they would work or not.”

Member of the public 2075

Precision Oncology – a natural evolution of healthcare

First example of Targeted Therapy: George Beatson
– Oopherectomy for Breast Cancer. *The Lancet July 18, 1896*



ON THE TREATMENT OF INOPERABLE CASES OF CARCINOMA OF THE MAMMA: SUGGESTIONS FOR A NEW METHOD OF TREATMENT, WITH ILLUSTRATIVE CASES.[1]

BY GEORGE THOMAS BEATSON, M.D. EDIN.,

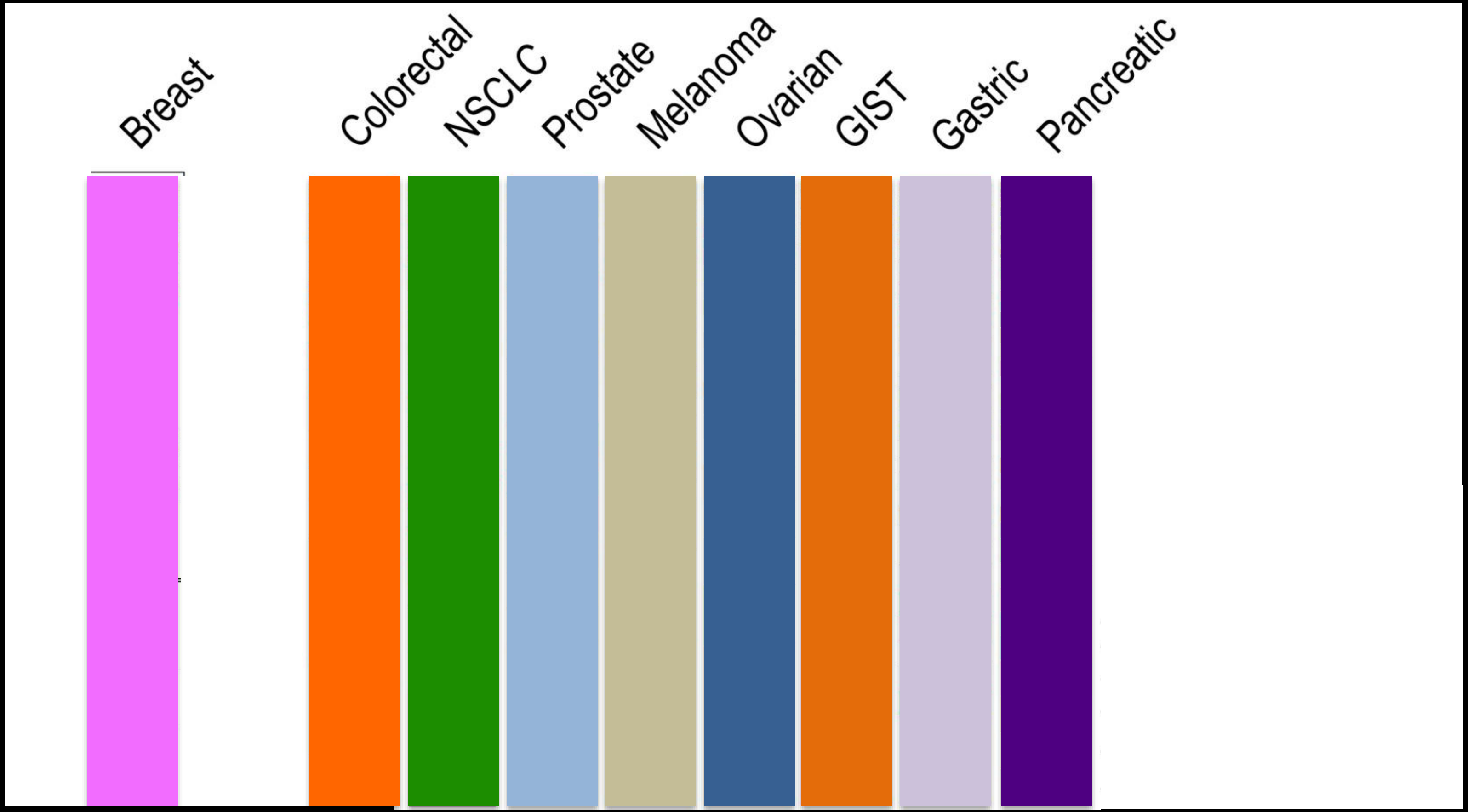
SURGEON TO THE GLASGOW CANCER HOSPITAL; ASSISTANT SURGEON GLASGOW WESTERN INFIRMARY; AND EXAMINER IN SURGERY TO THE UNIVERSITY OF EDINBURGH.

(*Concluded from page 107*).

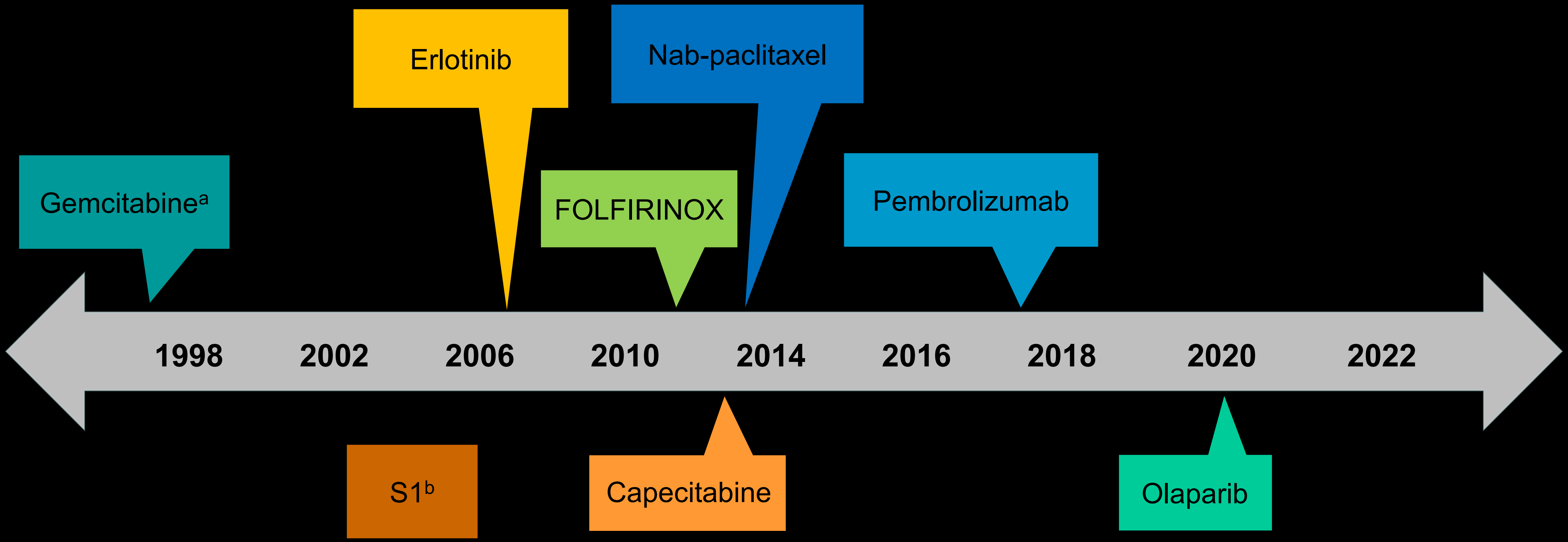
THE next case that I wish to bring under notice is that of a married woman aged forty years, with no family, who was admitted to the Glasgow Cancer Hospital on Sept. 2nd, 1895, suffering from a large tumour of the right mamma. It had

Most Successful
Targeted Therapy:
Tamoxifen (Developed
in the 1970's)

Cancer is complex



Treating pancreatic cancer:
Increasing availability of therapies with incremental benefit

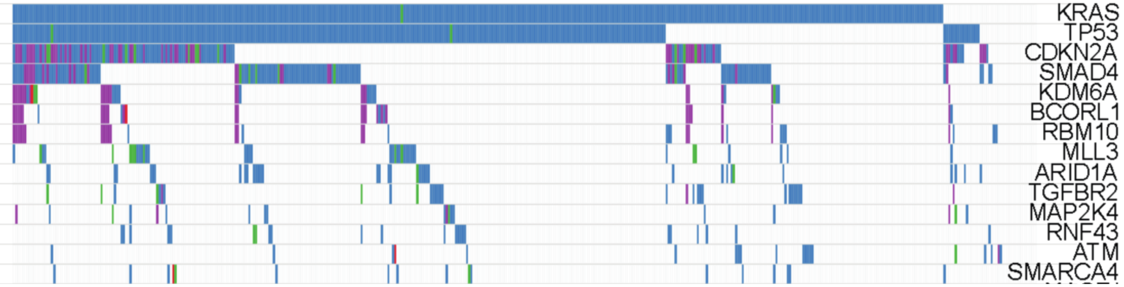


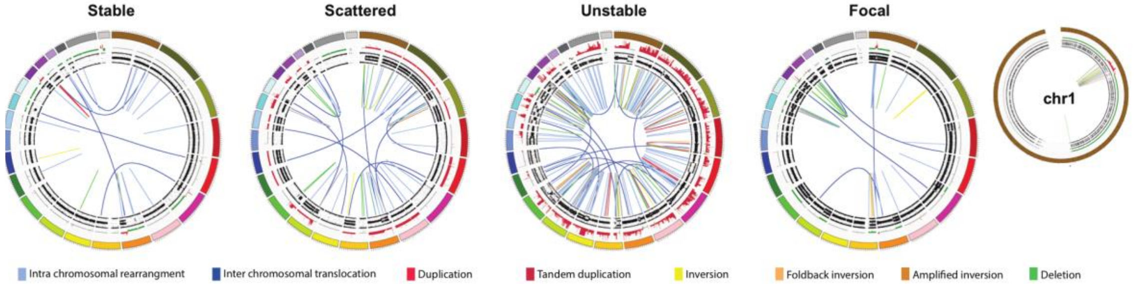
[a] Subsequently, gemcitabine-based combinations (eg, gemcitabine/capecitabine) also validated as treatment options in metastatic disease.[1,2] [b] Approved only in Japan.

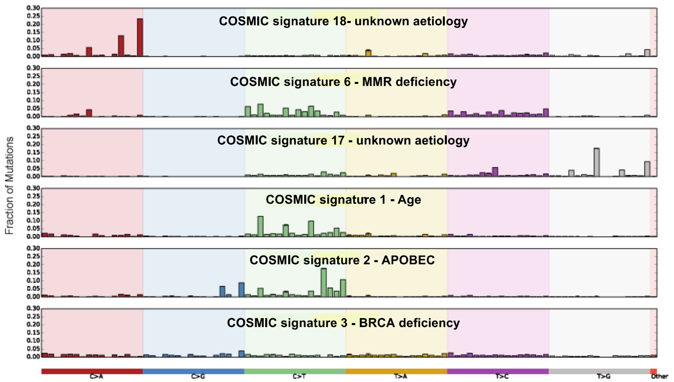
1. Herrmann R et al. *J Clin Oncol*. 2007;25:2212-2217. 2. Cunningham D et al. *J Clin Oncol*. 2009;27:5513-5518.

Modified Slide courtesy of Philip Philip

Key Molecular Mechanisms and Potential Novel Vulnerabilities (n = 457 multiplatform -omic analysis)







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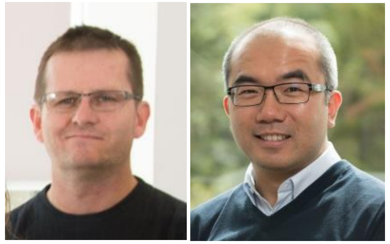


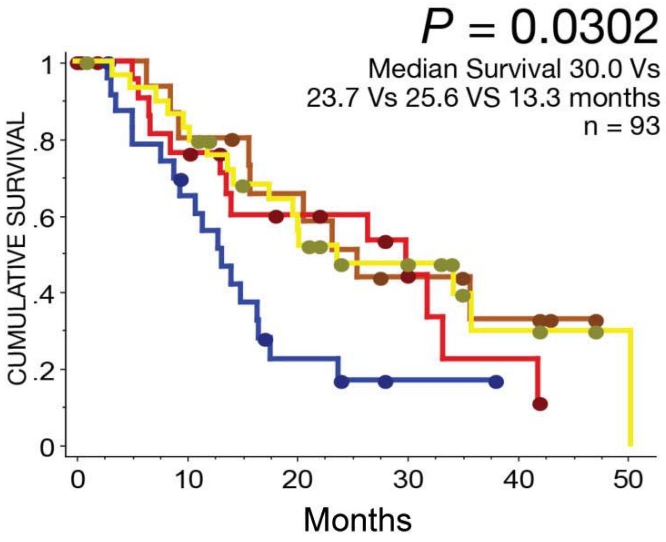
Peter Bailey
University of Glasgow

David Chang
University of Glasgow

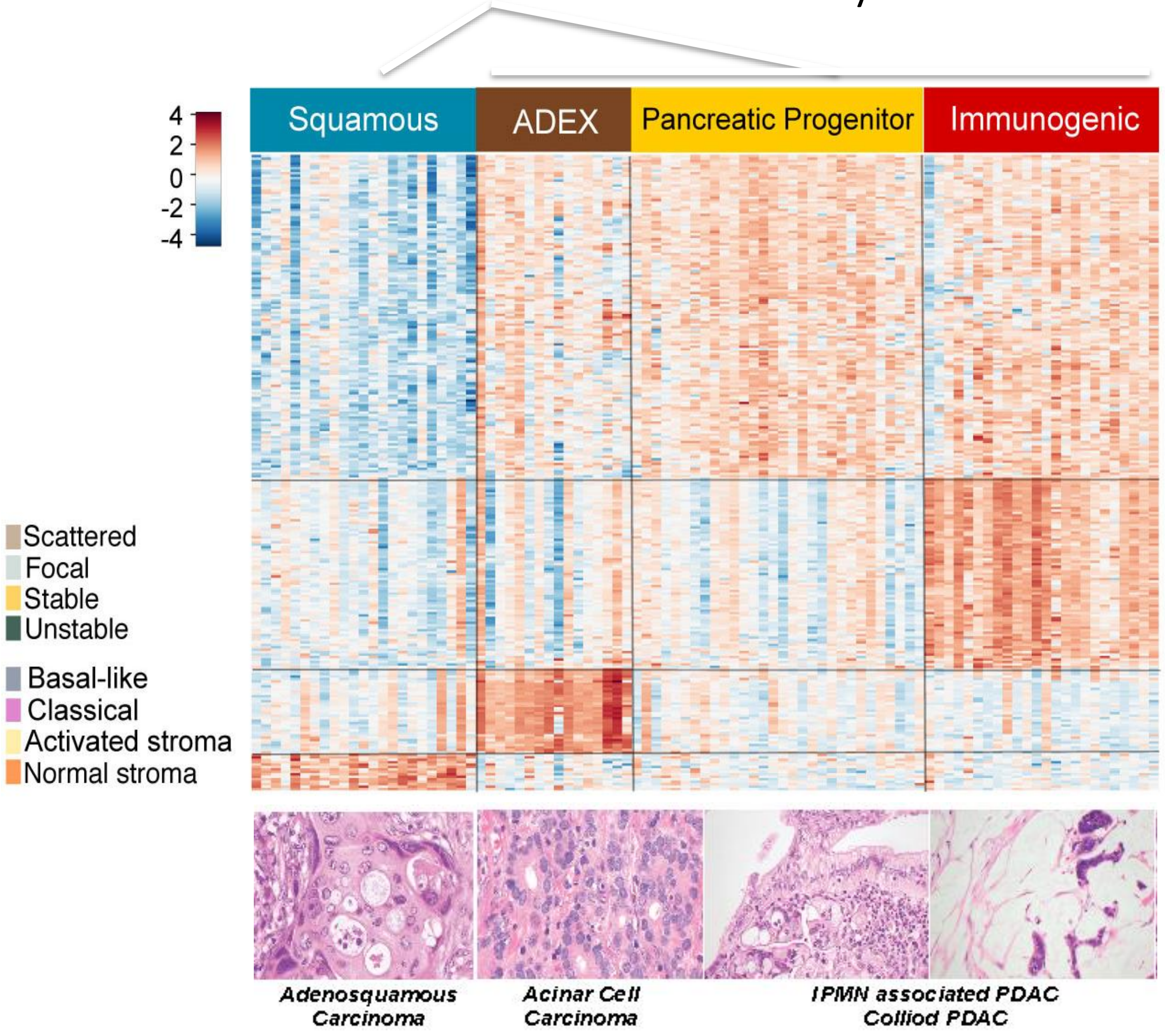
Hudson et al. Nature 2010
Biankin et al. Nature 2012
Perez-Mancera et al. Nature 2012
Mann et al. PNAS 2012
Alexandrov et al. Nature 2013
Nones et al. IJC 2013
Weissmueller et al. Cell 2014
Waddell et al. Nature 2015
Biankin et al. Nature 2015
Bailey et al. Nature 2016
Scarpa et al. Nature 2017
Feigin et al. Nature Genetics 2017
Balachandran et al. Nature 2017
TCGA Cancer Cell 2017

Transcriptomic Subtypes of PDAC

Bailey *et al. Nature* 2016

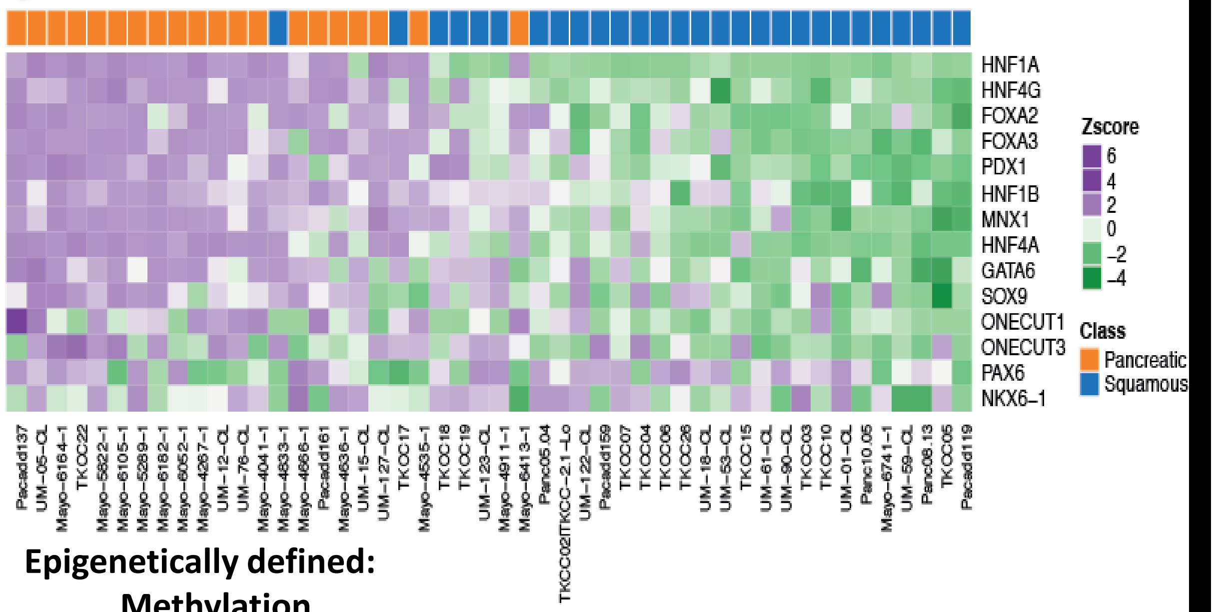


N = 96; Discovery RNAseq
N = 261; Validation expression array



Pancreatic Cancer Phylotranscriptomic Tree

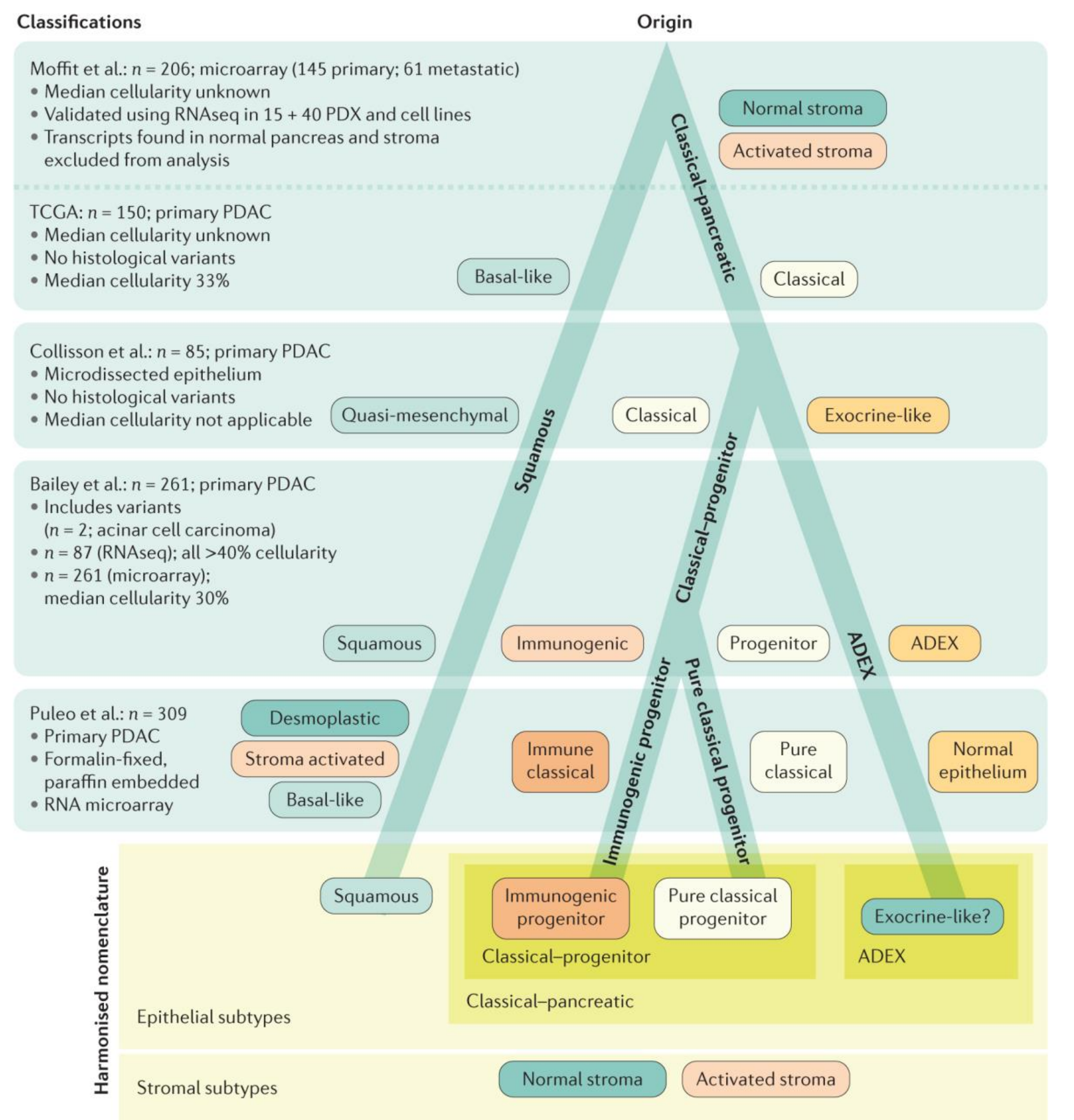
ADEX (x3) **Pancreatic (Classical)** **Squamous**



Epigenetically defined:

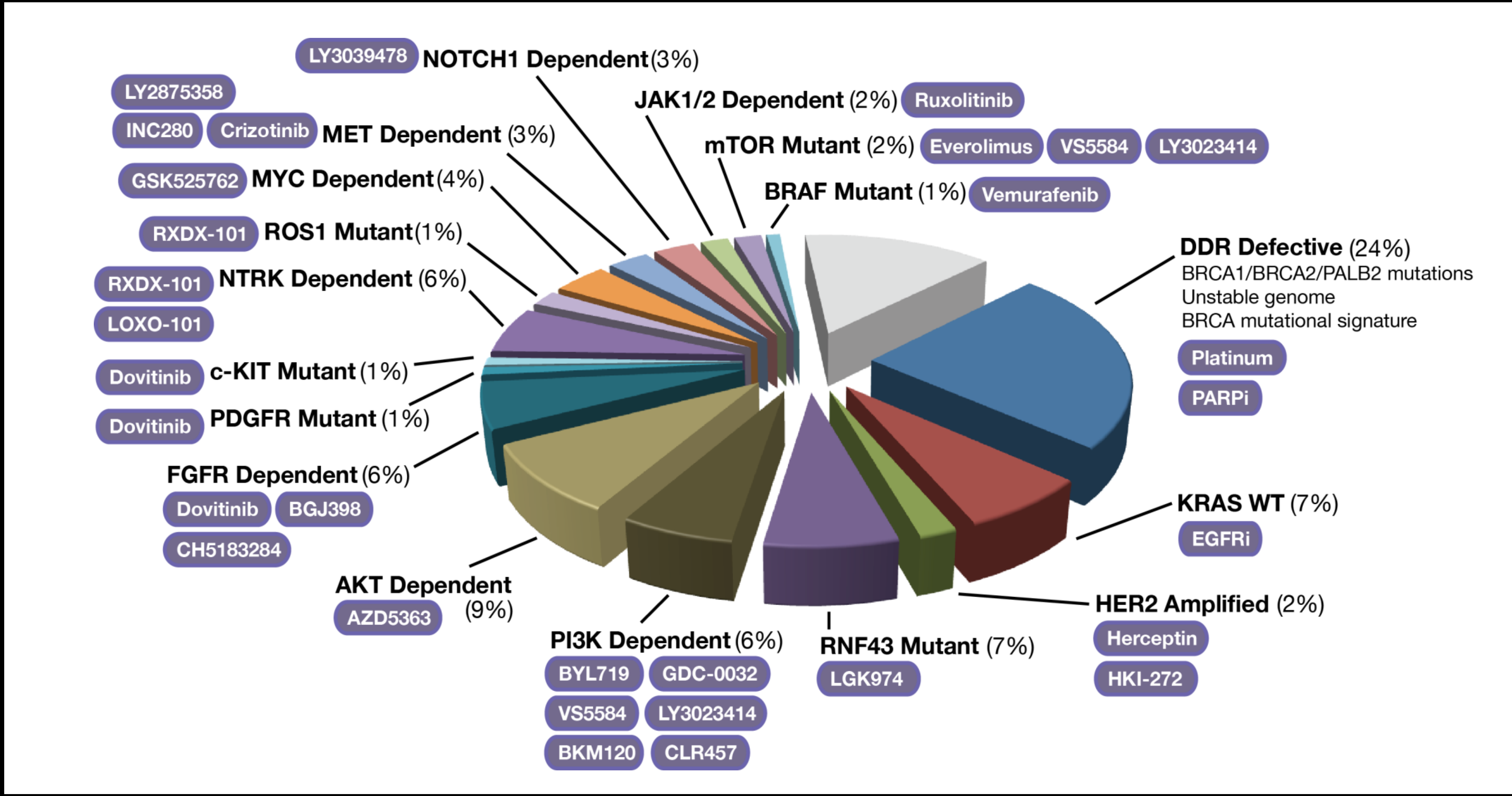
Methylation

Mutations in Chromatin Modifier Genes

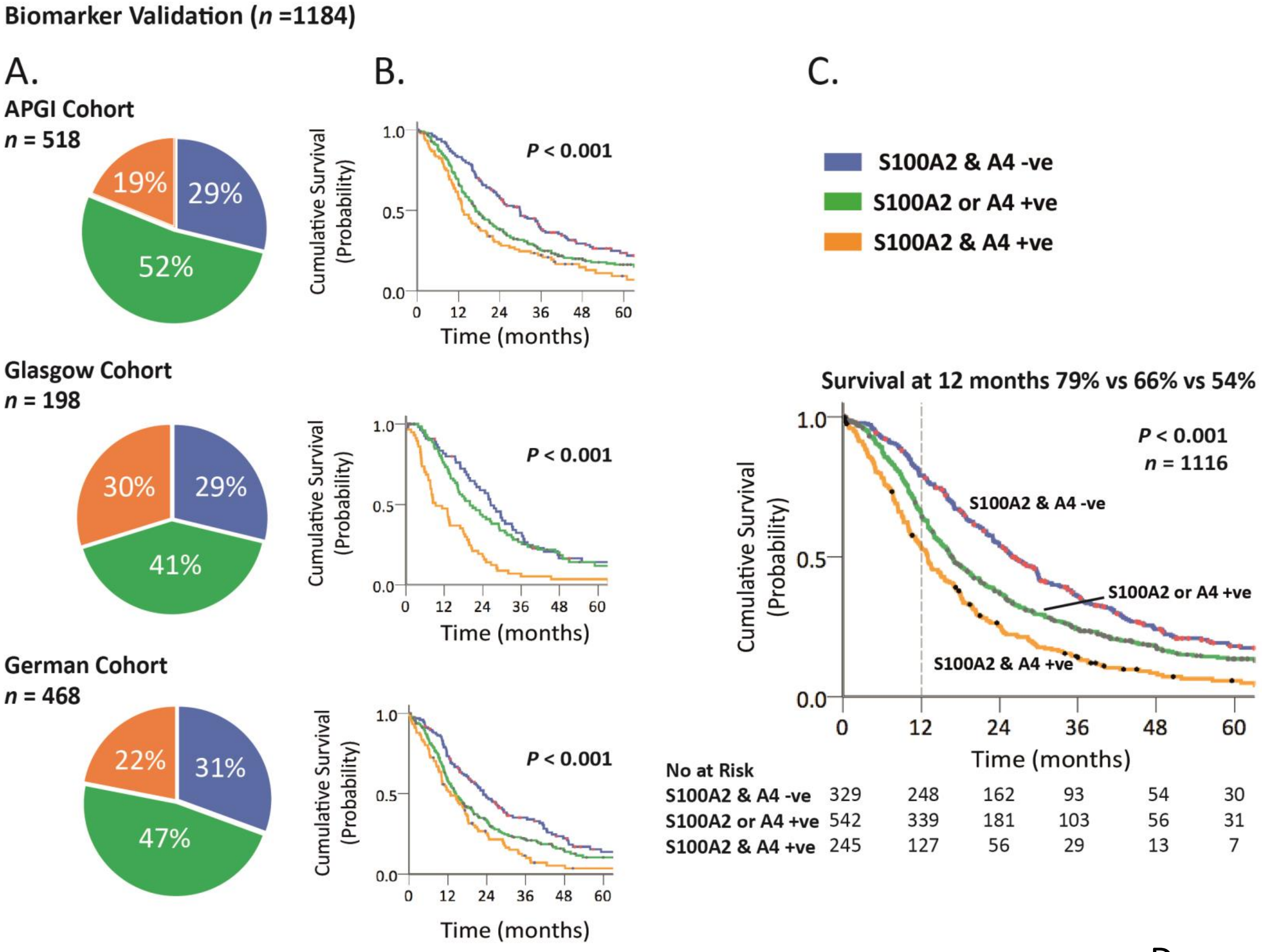


Molecular subtypes of pancreatic cancer.
Collisson EA, Bailey P, Chang DK, Biankin AV.
Nat Rev Gastroenterol Hepatol. 2019;16:207-220.

Pancreatic Cancer “Druggable Genome” (primary resectable)

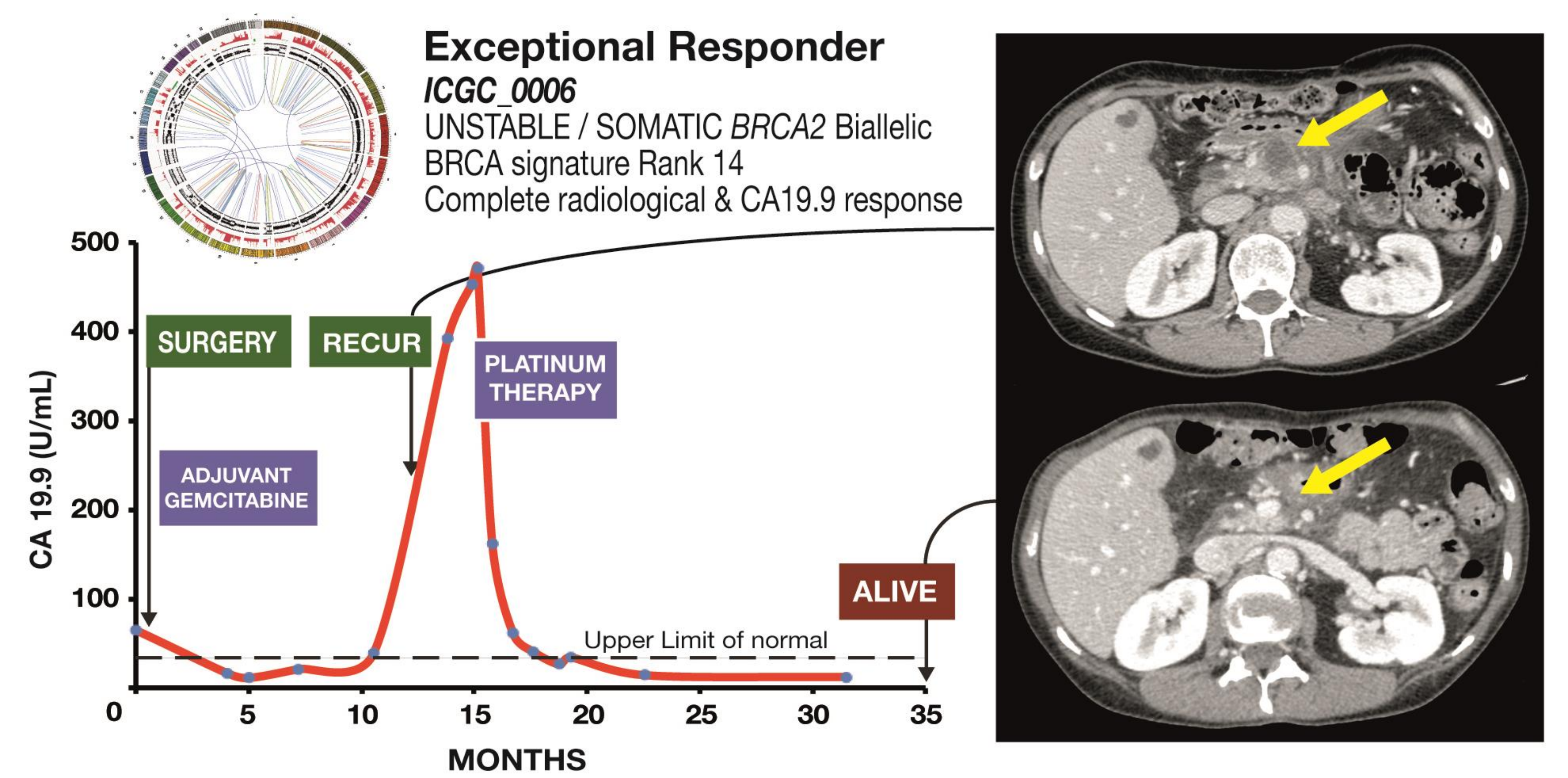


High S100A2/A4 = poor response to surgery



Dreyer et al. 2018

Pancreatic Cancer (64 year old female)



Sustainability

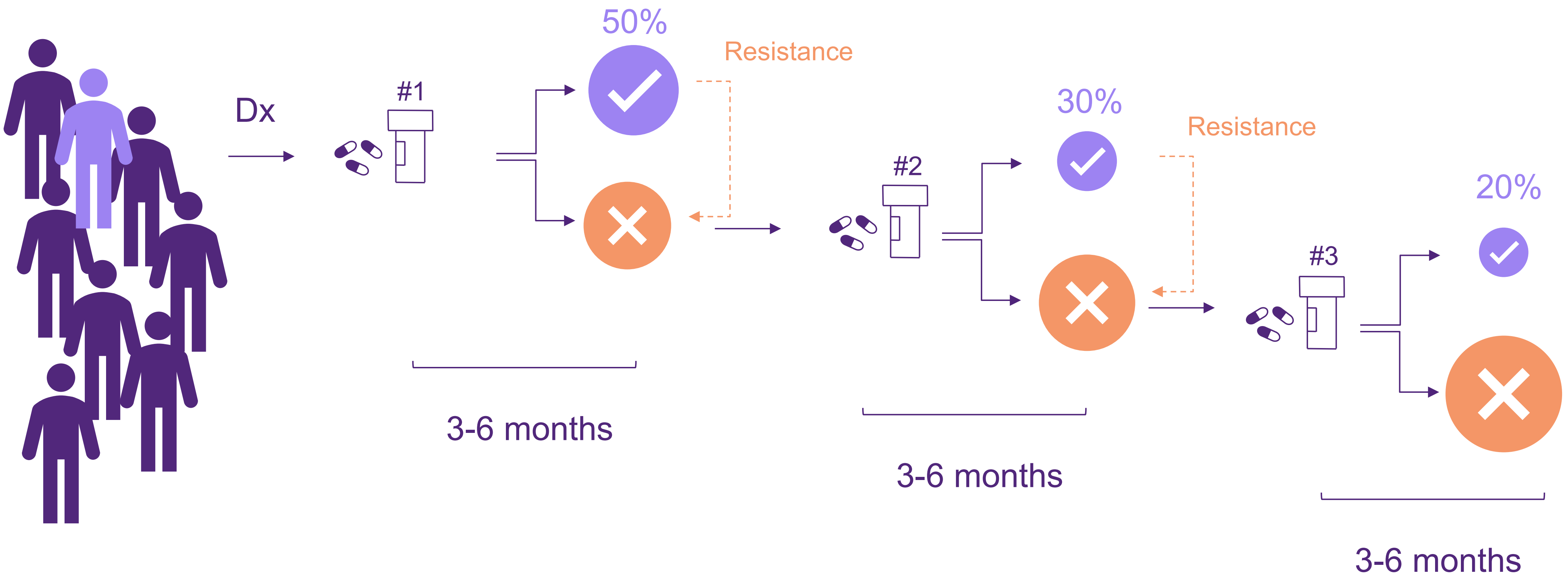
Better outcomes

Quality of life

Current cancer treatment pathway

Trial and error based on average response rates

Cannot predict ahead of time if a patient is going to respond to a treatment or have best possible outcome



Current cancer treatment pathway

Trial and error based on average response rates

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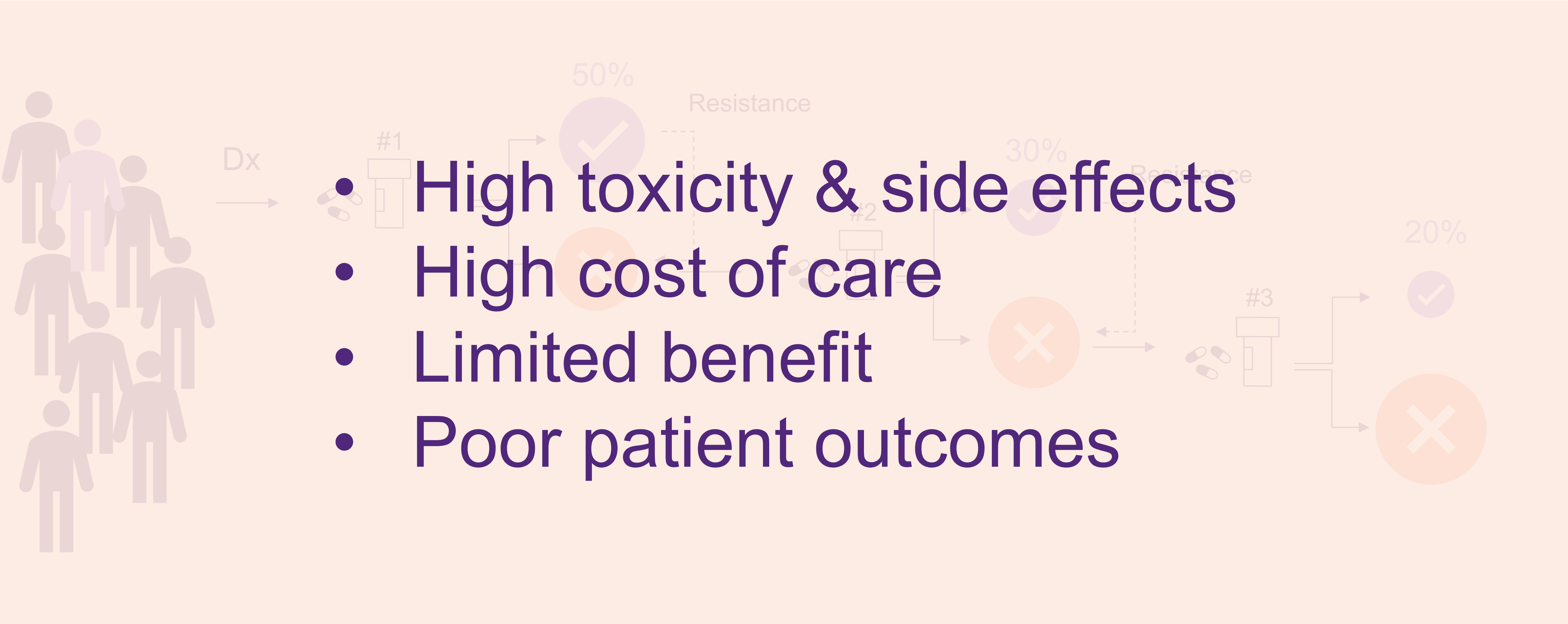

- High toxicity & side effects
- High cost of care
- Limited benefit
- Poor patient outcomes

Why individual patient response is important

Treatment Options	Overall Response Rates	Mrs A.S.	Mr J.W.	Ms P.B.	Mr F.A.B.	Prof A.V.B
Treatment A	55%	5%	95%	22%	9%	5%
Treatment B	50%	18%	11%	77%	9%	2%
Treatment C	42%	85%	2%	16%	6%	1%
Treatment D	32%	22%	29%	9%	2%	6%
Clinical Trial	?	45%	30%	17%	63%	8%
						?

Increasing value of better decisions

Current oncology drug development process is expensive, long and has high attrition rates

Phase	Failure Rate	% Dev Cost	Cost ($, m)	Cost of failure ($, m)	Attrition ($, m)
Preclinical	60%	7%	$140	$11	$84
Phase 1	35%	15%	$300	$59	$105
Phase 2	50%	21%	$420	$126	$210
Phase 3	40%	26%	$520	$312	$208
Overall	**99.5%**	**86%**	**$2,000**	**-**	**$927**

Source: Adapted from Sun et al (2022), Wouters et al (2020)

DATA, DATA, DATA …

Real World Data
Prospective Data
Clinical Trial Data
Molecular Data

GLASGOW PRECISION ONCOLOGY LABORATORY

The next generation of cancer tests

The Glasgow Cancer Test

Susie Cooke
Deputy Director
Medical Genomics
and Informatics

Andrew Biankin
Director
Therapeutic and Diagnostic
Development

Philip Beer
Clinical Cancer Genomics

University of Glasgow
ARGO Accelerating Research in Genomic Oncology
International Cancer Genome Consortium
NHS
wellcome sanger institute

Clinical Trial Platforms

CANCER RESEARCH UK | PRECISION PANC

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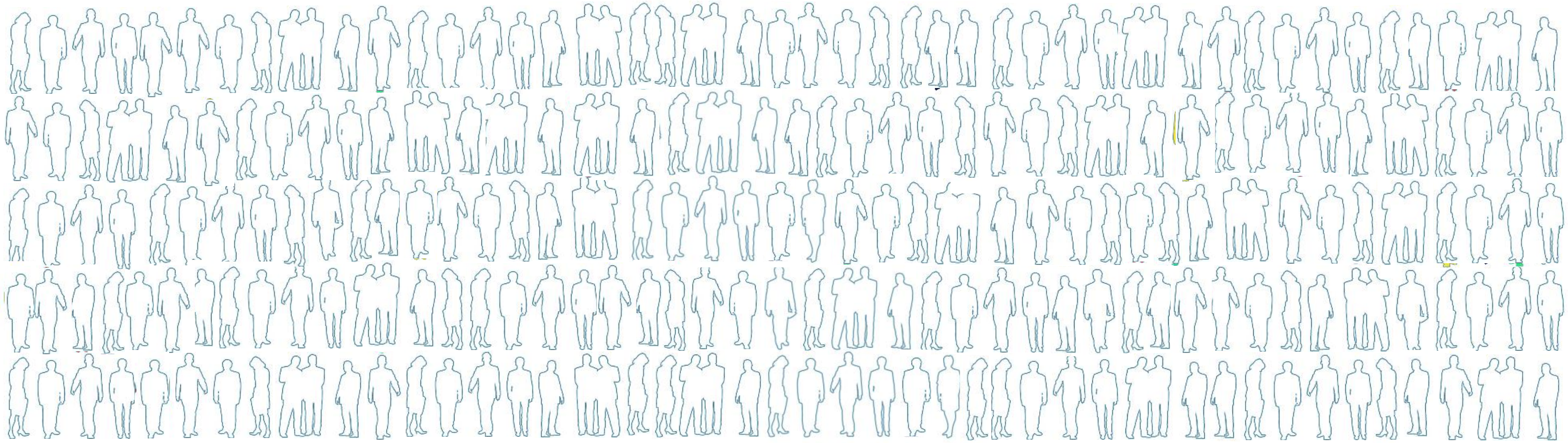
Precision-Panc

Not all cancers are the same. Precision medicine is about tailoring treatments to an individual's cancer. Precision-Panc Clinical Trials are delivered through the NHS and match people with pancreatic cancer to the trial most likely to work for them.

From single to multiple gene tests



TEST FOR MANY GENES AT ONCE

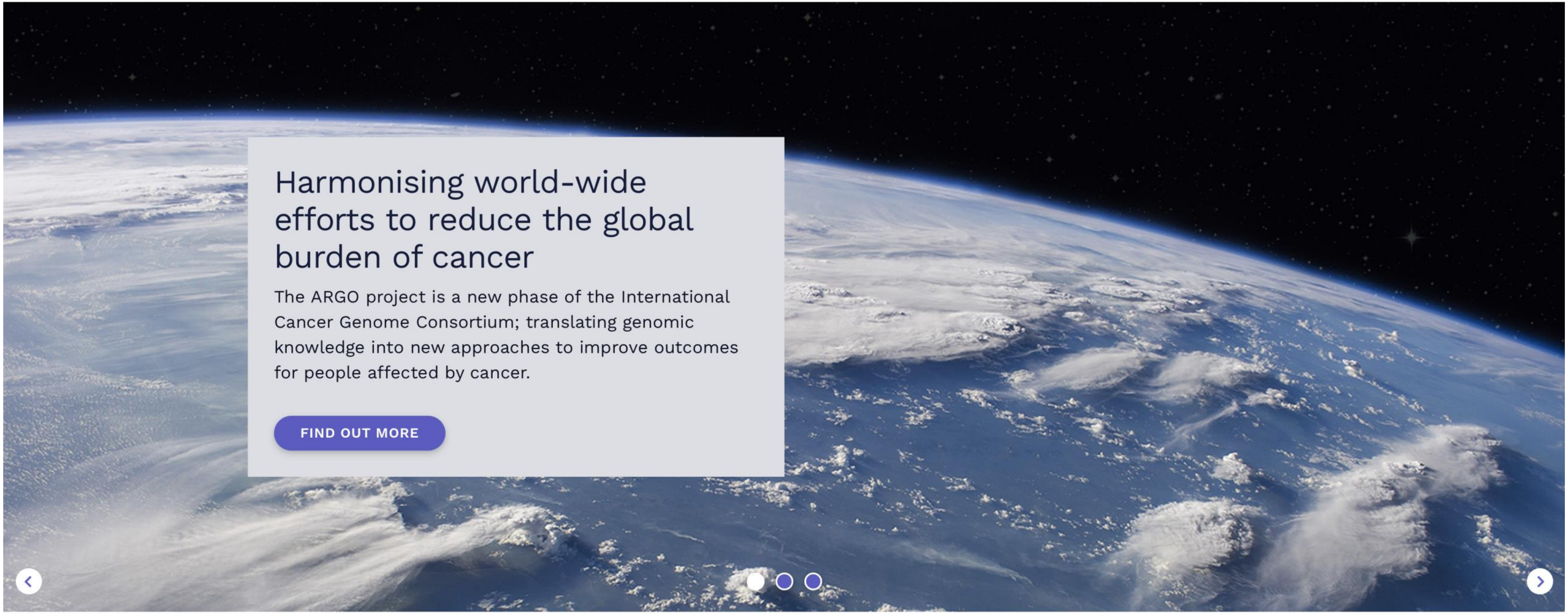
Novel Targets and Drugs

Trial of Drug A

Trial of Drug B

Trial of Drug C

Trial of Drug D



Harmonising world-wide efforts to reduce the global burden of cancer

The ARGO project is a new phase of the International Cancer Genome Consortium; translating genomic knowledge into new approaches to improve outcomes for people affected by cancer.

FIND OUT MORE

Accelerating Research in Genomic Oncology

ICGC ARGO will uniformly analyze specimens from 100,000 cancer patients with high quality clinical data to address outstanding questions that are vital to our quest to defeat cancer.

26 Programs, 13 countries, 63,116 donors committed

Weill Cornell Medicine Englander Institute for Precision Medicine
Weill Cornell Precision Medicine Program (USA, multiple cancers)

Swiss Oncology and Cancer Immunology Breakthrough Platform (Switzerland, multiple cancers)

SAMSUNG MEDICAL CENTER
Personalized Genomic Characterisatiopn of Korean Lung Cancers (Korea)

OCCAMS
Oesophageal Cancer Clinical and Molecular Stratification (UK)

PBCP
Personalised Breast Cancer Program (United Kingdom)

SNUH 서울대학교병원
Korean Multiple Myeloma Precision Medicine Project (Korea)

Pan Prostate Cancer Group (United Kingdom)

국립암센터 NATIONAL CANCER CENTER
Korean Rare Cancers Project (Korea)

OICR
Pancreatic Cancer Harmonized "Omics" analysis for Personalized Treatment (Canada)

BC Cancer personalised OncoGenomics Program (Canada, multiple cancers)

Papillary Thyroid Cancer Project (Saudi Arabia)

CCGC
Chinese Cancer Genome Consortium (China, colorectal cancer)

CANCER RESEARCH UK | GRAND CHALLENGE MUTOGRAPHS
Mutographs Study (UK, France, multiple cancers)

PRECISION PANC
Precision Panc (UK, pancreatic cancer)

P1000
1000 Polyethnic Study (USA, multiple cancers)

UMG G-CCC
European Peripheral T Cell Lymphoma Study (Germany)

China Diffuse Gastric Cancer Study (China)

GUSTAVE ROUSSY
Non-Melanoma Skin Cancer Project, (FRANCE)

International Oesophageal Cancer Genome
Oesophageal Squamous Cell Carcinoma Study (China)

AMED
Genomic Medicine for Asia Prevalent Cancers (Japan, multiple cancers)

Profiling Orphan Neoplasms for Treatment (Italy, multiple cancers)

MONSTAR
Monstar Project (multiple cancers, Japan)

The Chinese University of Hong Kong
Hong Kong Brain Metastasis Study, Colorectal and Lung Cancers (Chinese University of HK)

UCL
OLIGO Clinical Trial, (multiple cancers) University College London (UK)

More details on member programs: https://www.icgc-argo.org/page/89/project-list

THE LANCET Oncology

ARGO Accelerating Research in Genomic Oncology

International Cancer Genome Consortium

COMMENT | VOLUME 24, ISSUE 2, P123-125, FEBRUARY 2023

Accelerating cancer omics and precision oncology in health care and research: a *Lancet Oncology* Commission

Raffaella Casolino • Amber L Johns • Melanie Courtot • Rita T Lawlor • Francesco De Lorenzo • Denis Horgan • Joaquin Mateo • Nicola Normanno • Mark Rubin • Lincoln Stein • Vivek Subbiah • Benedikt C Westphalen • Mark Lawler • Keunchil Park • Sandra Perdomo • Takayuki Yoshino • Jianmin Wu • Andrew V Biankin • on behalf of the Lancet Oncology Commission on Cancer Omics and Precision Oncology • Show less

Published: February, 2023 • DOI: https://doi.org/10.1016/S1470-2045(23)00007-4 • Check for updates

genomeweb

Business & Policy | Technology | Research | Diagnostics | Disease Areas | Applied M

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International Commission to Expand Molecular Testing Access for Cancer Patients

Jan 31, 2023 | staff reporter

NEW YORK – The University of Glasgow said on Tuesday that it will lead a new commission dedicated to improving global access to diagnostic tests and personalized cancer care.

The Lancet Oncology Commission, comprising an international team of experts, will study obstacles to widespread adoption of molecular testing for cancer diagnosis and treatment and generate practical solutions for expanding access and reducing global health inequities. The commission will analyze how health and genomic data are collected and shared in order to develop best practices and facilitate drug development.

University of Glasgow to lead internationa effort to advance cancer care and researcl

Hippocratic Post | 31st January 2023 | CANCER, GLOBAL REACH | No Comments

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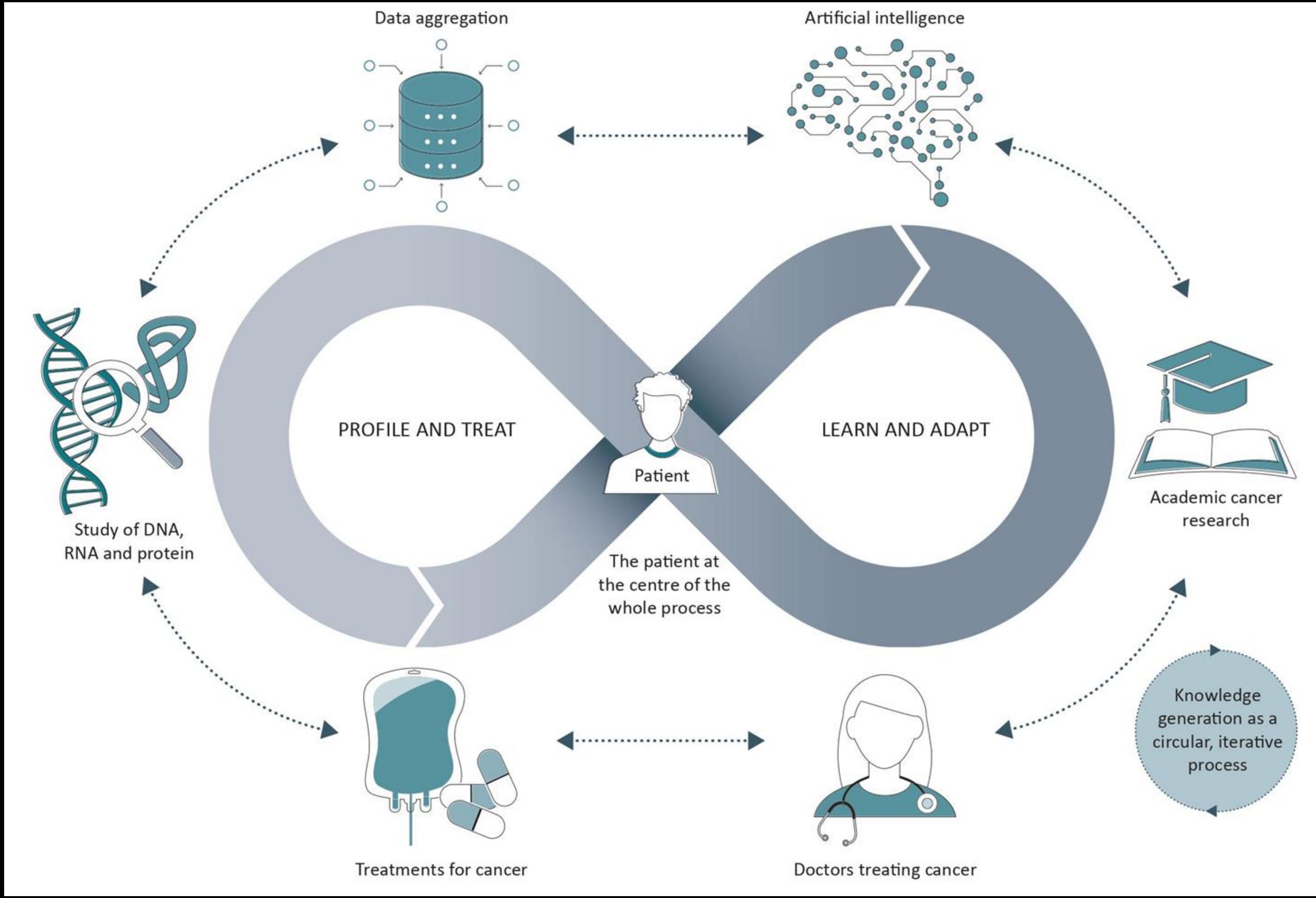
Bridging the implementation gap

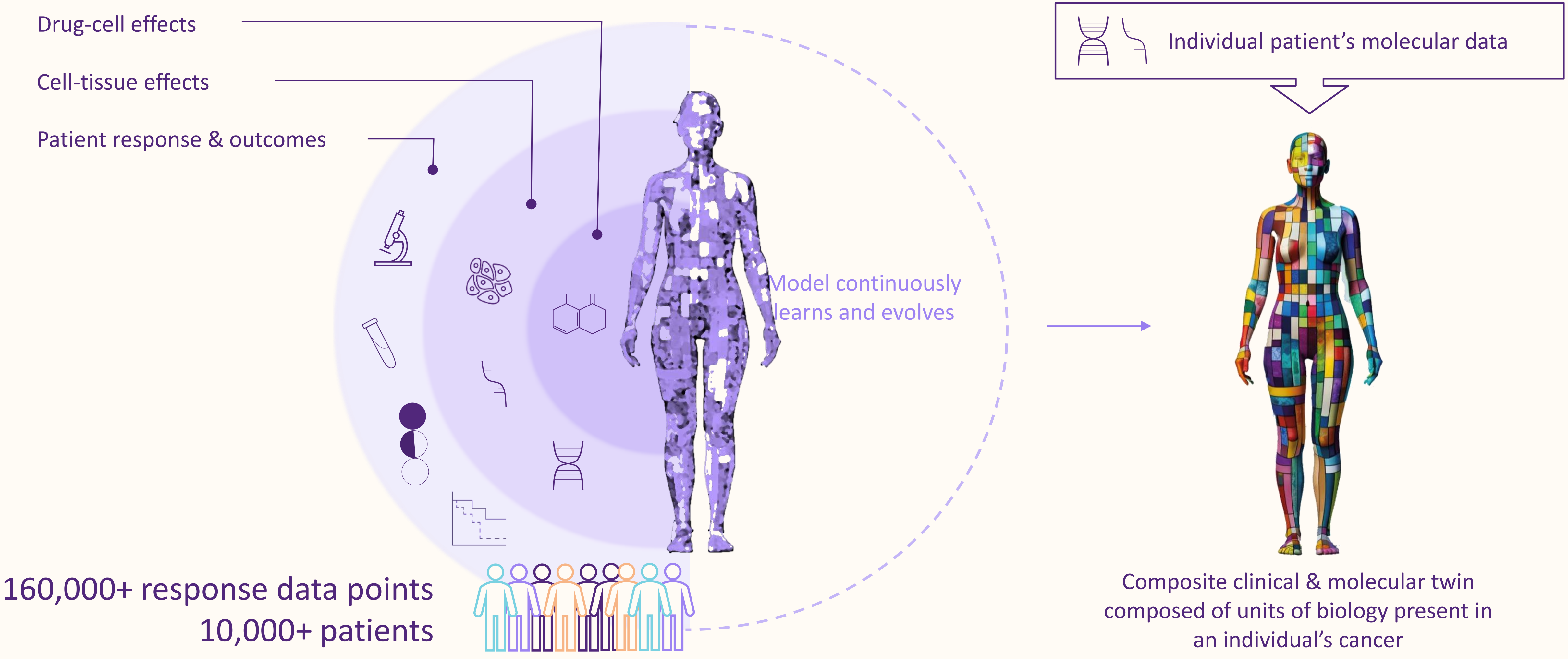
Taken from:
Beer and Biankin, Bridging the implementation gap: delivering complex genomic analysis for routine cancer care. Annals of Oncology, *in press*.



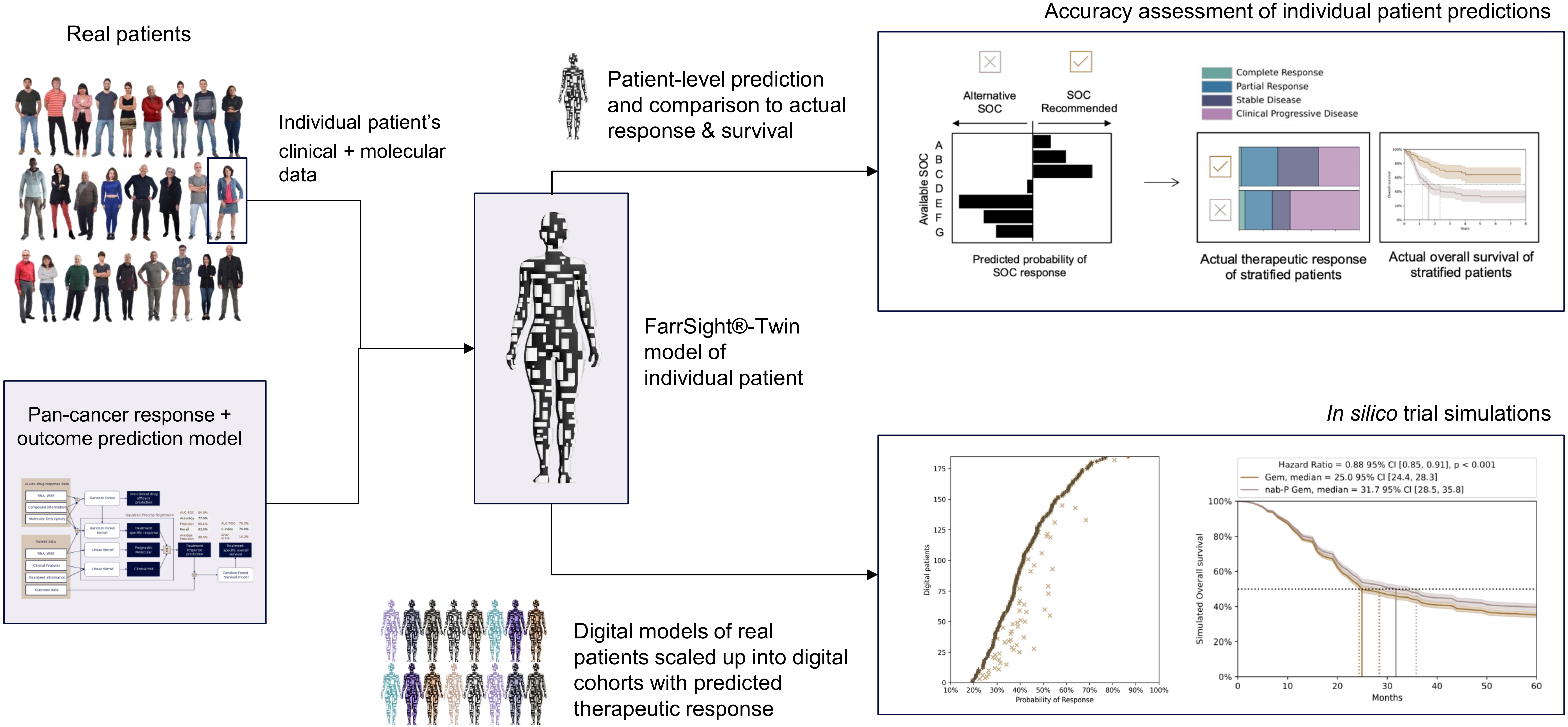
AI in Healthcare and Therapeutic Development

Digital Twins of Cancer Patients

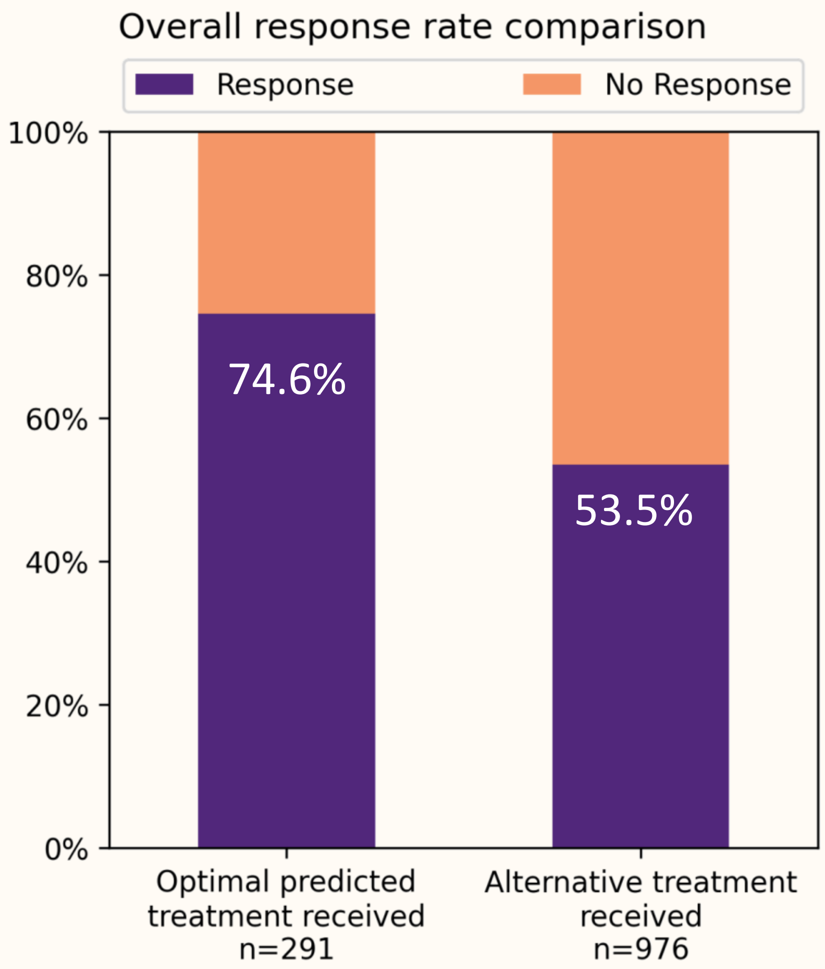
FarrSight®-Twin: the next-generation of digital twin to predict individualised therapeutic response and outcome



FarrSight®-Twins of each and every individual patient



FarrSight®-Twin-predicted treatment = better overall response

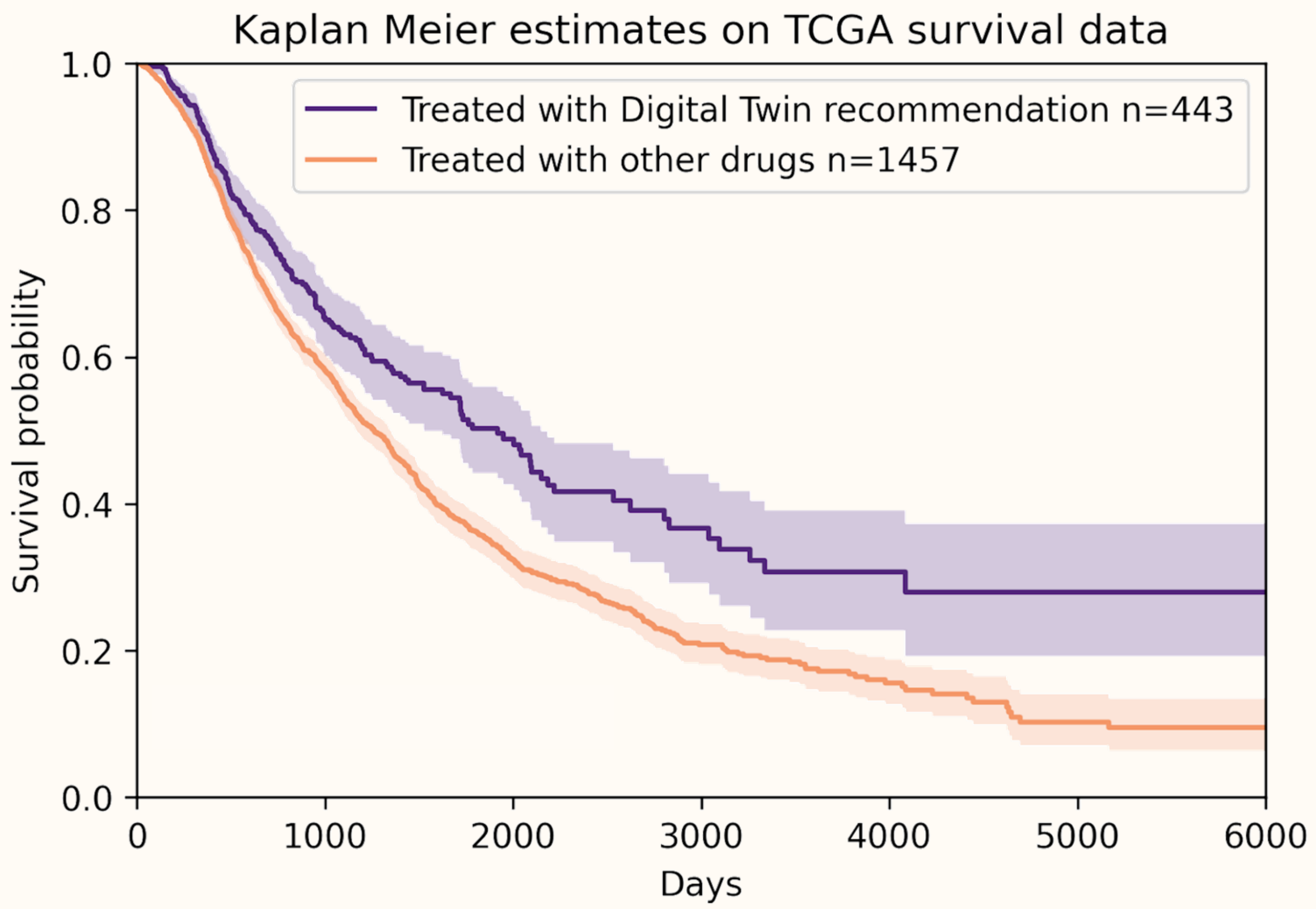


Odds ratio = 2.55

Fisher exact $p=1 \times 10^{-10}$

Significantly higher overall response rates in patients who received treatment recommended by FarrSight®-Twin compared to patients who didn't receive the treatment predicted to be optimal

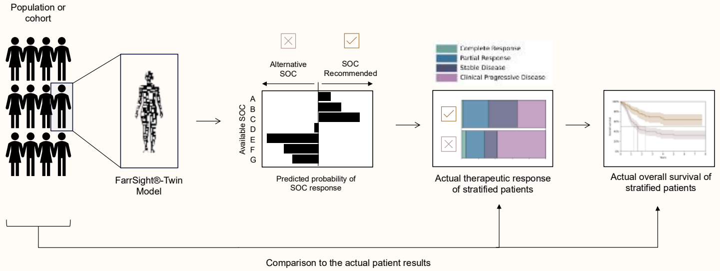
Optimal FarrSight®-Twin-predicted treatment = better survival

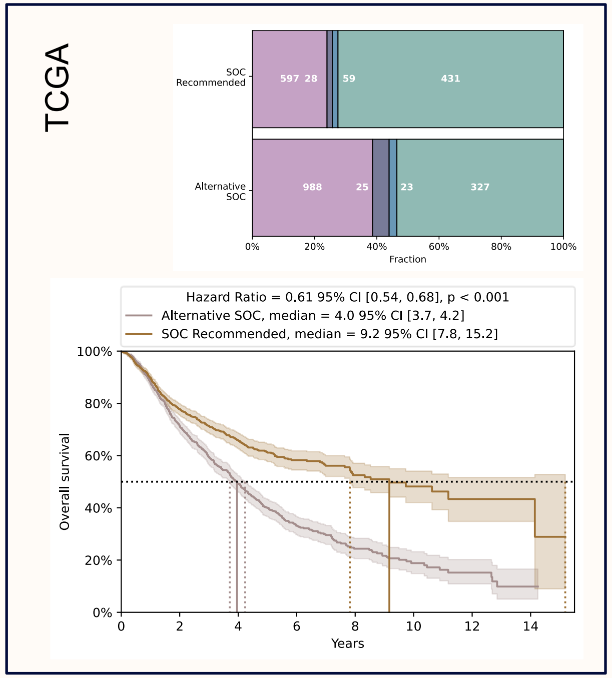


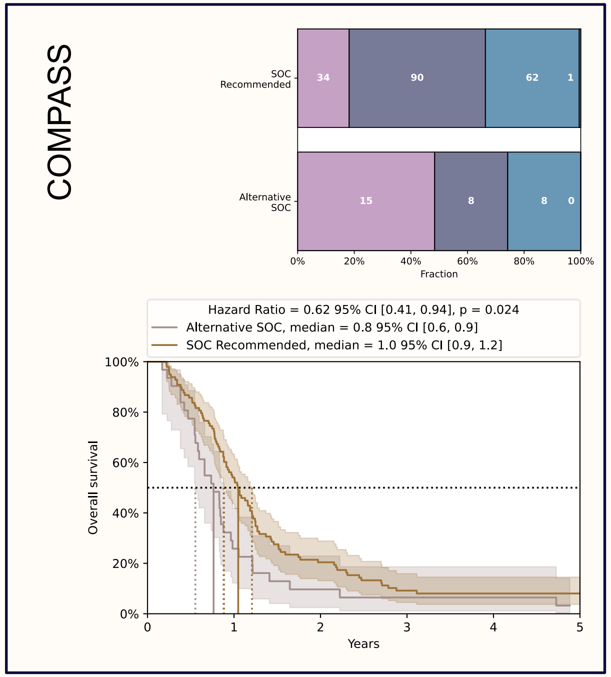
Log-rank p-value=2.5E-6

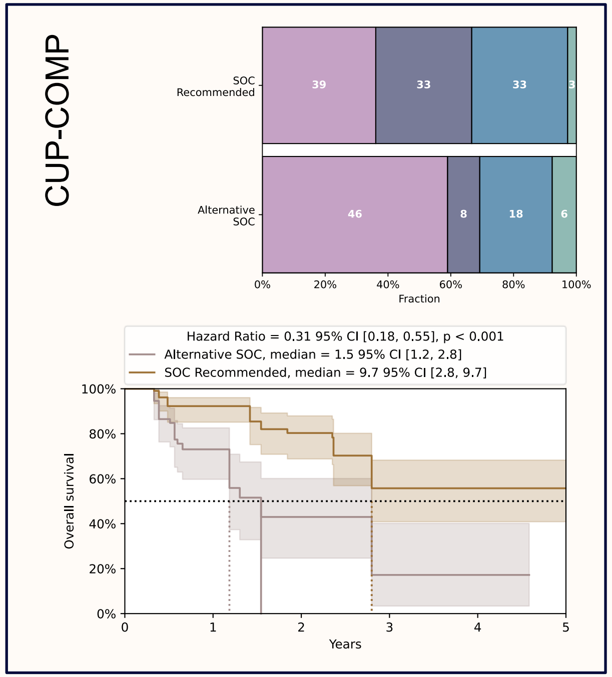
Significantly better overall survival in patients who received FarrSight®-Twin-predicted therapy compared to those who did not

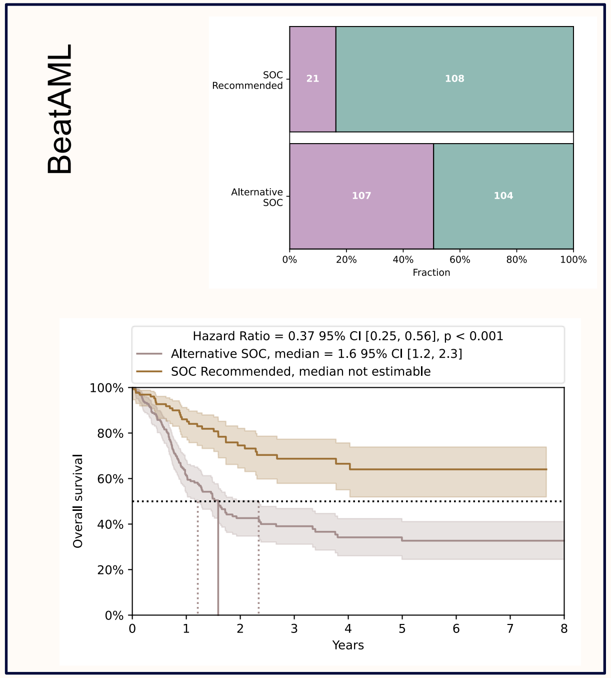
FarrSight®-Twin predicts optimal therapy across datasets



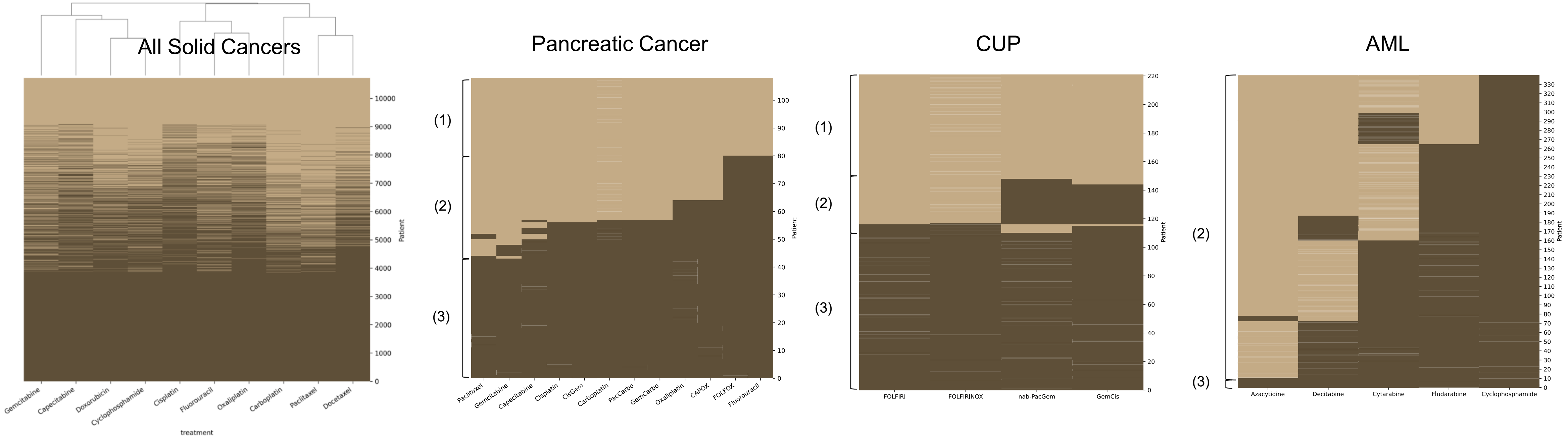






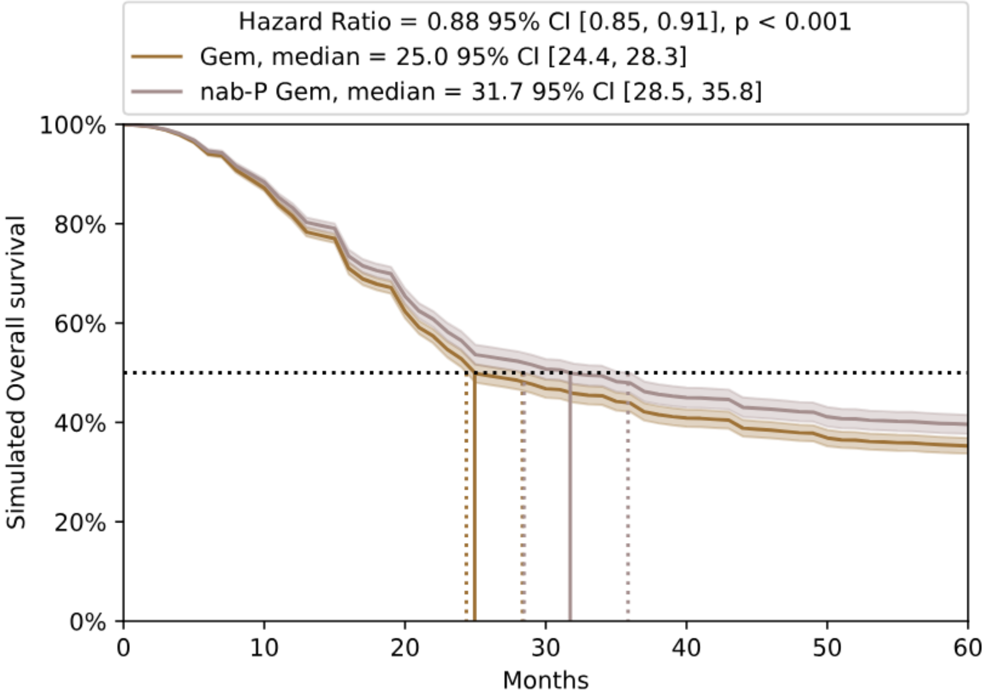


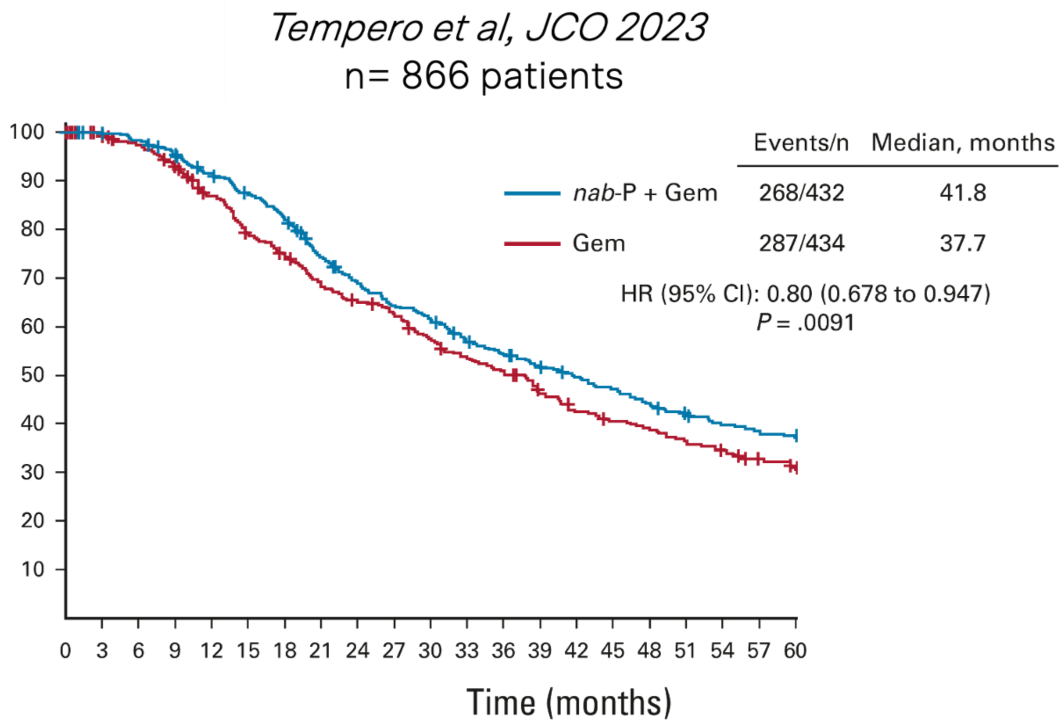
Chemotherapy Response (all chemotherapies)



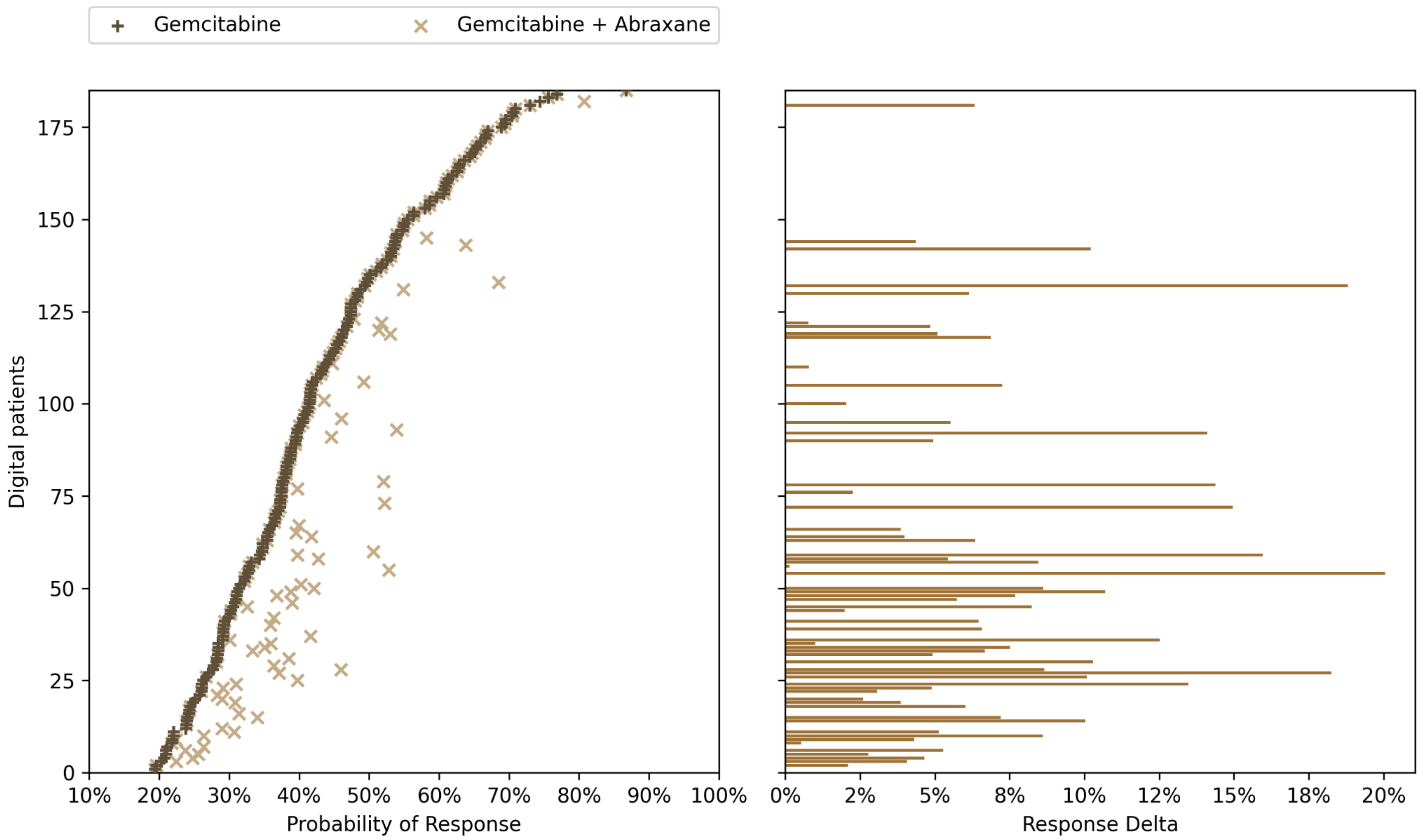
Predicted responses across multiple cancer types

Application in research clinical trials: RWD for trial simulation





FarrSight®-Twin scaling to cohorts captures real-world response heterogeneity



Transforming Healthcare:

Use current treatments better
Predicting clinical trial and “off label” options

Select drugs for development
Select indications
Understand MOA and Biology
Biomarker discovery and development
Synthetic controls
Single arm combination studies
Clinical trial design

Transforming Healthcare (Challenges):

Adoption:

What evidence is required?

Compared to What?

Regulators and Payers

Integration with current systems

Inertia

University of Glasgow

Thank you